

Energy Portfolio Risk Modelling in R

Overview

This project develops and evaluates a compact market-risk framework for a synthetic multi-asset energy portfolio using real historical market data.

The objective is to estimate short-horizon portfolio losses using different Value-at-Risk methodologies, assess tail risk through Expected Shortfall, validate the models using rolling backtests, and evaluate portfolio behaviour under severe stress scenarios.

The project focuses on three energy-market risk factors:

- German day-ahead electricity prices
- Henry Hub natural gas prices
- Brent crude oil prices

The analysis is implemented in **R**.

Data

The sample covers approximately:

January 2019 to December 2025

The final merged dataset contains daily observations available across all three markets.

The underlying series are:

- German day-ahead electricity prices in EUR/MWh
- Henry Hub natural gas spot prices in USD/MMBtu
- Brent crude oil spot prices in USD/barrel

The datasets are aligned on common observation dates. Missing trading days are not forward-filled, since this would artificially introduce zero price changes.

Risk-factor construction

The portfolio is modelled using daily absolute price changes:

$$P_t = P_t - P_{t-1}$$

This approach is used for all three commodities.

Absolute price changes are particularly appropriate for electricity because power prices can be zero or negative, making conventional logarithmic returns unsuitable.

Portfolio construction

A synthetic long-only energy portfolio is constructed with approximate initial exposures of:

- €4 million to German power
- €3 million to Henry Hub gas
- €3 million to Brent crude oil

The portfolio therefore has an initial gross exposure of approximately:

€10

Positions are represented through fixed physical quantities.

For each commodity:

$$PnL_{i,t} = q_i \Delta P_{i,t}$$

where q_i is the fixed quantity of commodity i.

Portfolio PnL is therefore:

$$PnL_t = q_P \Delta P_{P,t} + q_G \Delta P_{P,t} + q_B \Delta P_{P,t}$$

A fixed USD/EUR conversion factor is used for Henry Hub and Brent exposures. This deliberately removes FX risk from the analysis so the project remains focused on commodity-market risk.

Exploratory risk characteristics

The three risk factors exhibit very weak contemporaneous correlations.

Approximate pairwise correlations are only around 0.02–0.04.

However, all three distributions exhibit substantial non-normality.

Estimated kurtosis:

- German power: approximately 12
- Henry Hub gas: approximately 269
- Brent crude: approximately 10

For comparison, a normally distributed variable has kurtosis equal to 3.

Henry Hub in particular contains several extreme observations, while German power exhibits large positive and negative daily price movements.

These characteristics motivate comparing a Gaussian VaR model against a non-parametric historical simulation approach.

Parametric Value-at-Risk

The first model assumes portfolio P&L is normally distributed.

The model estimates:

$$\mu PnL$$

and

$$\sigma PnL$$

from historical portfolio P&L.

Value-at-Risk is then obtained from the relevant quantile of the normal distribution.

The estimated risks are approximately:

Confidence level	VaR	Expected Shortfall
95%	€3.72m	€4.66m
99%	€5.26m	€6.02m

The parametric model provides a transparent benchmark but relies heavily on the assumption of normally distributed P&L.

Historical Simulation

Historical Simulation VaR is estimated directly from the empirical portfolio P&L distribution.

Unlike the parametric model, it does not impose a normal distribution and therefore preserves historically observed skewness and fat tails.

Estimated risks are approximately:

Confidence level	VaR	Expected Shortfall
95%	€3.14m	€5.84m
99%	€7.53m	€10.85m

The difference between the models becomes particularly large in the extreme tail.

At the 95% confidence level, the Gaussian model produces the larger VaR estimate.

At the 99% confidence level, however, Historical Simulation produces substantially larger VaR and Expected Shortfall estimates.

This reflects the heavy-tailed empirical distribution of energy-market P&L.

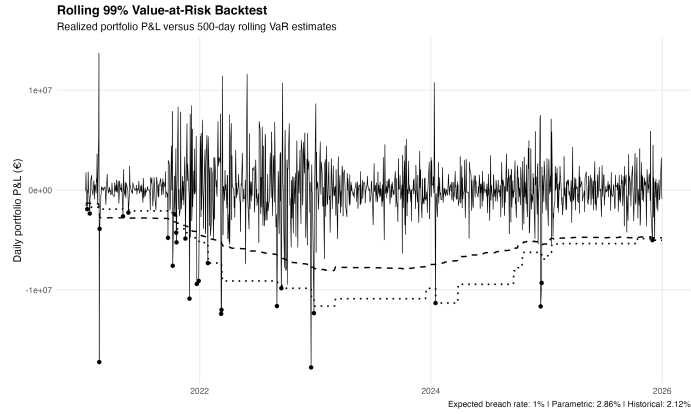


Figure 1: VaR Backtest

Rolling VaR backtesting

Both VaR models are evaluated using a rolling 500-observation estimation window.

For each date:

1. The previous 500 observations are used to estimate the model.
2. A one-day-ahead 99% VaR estimate is produced.
3. The predicted VaR threshold is compared with the realized portfolio P&L.
4. A breach is recorded whenever the realized loss exceeds predicted VaR.

The backtest contains 1,224 out-of-sample observations.

For a correctly calibrated 99% VaR model, approximately 1% of observations should exceed the VaR threshold.

Backtest results

Model	Breaches	Breach rate	Expected rate
Parametric	35	2.86%	1%
Historical	26	2.12%	1%

Both models therefore underestimate realized tail risk.

Historical Simulation performs better than the Gaussian model, but its breach rate remains more than twice the expected level.

Kupiec coverage test

The Kupiec unconditional coverage test is used to formally evaluate whether the realized frequency of VaR breaches is consistent with the nominal 1% probability.

Results:

- Parametric VaR: $p \approx 9.6 * 10^{-8}$
- Historical VaR: $p \approx 0.00056$

The null hypothesis of correct unconditional coverage is rejected for both models.

This indicates that neither model provides adequately calibrated 99% tail-risk forecasts over the evaluation period.

Stress testing

VaR is complemented with explicit stress scenarios.

Joint empirical 1% downside scenario

Each risk factor is simultaneously subjected to its empirical 1% downside shock.

The resulting portfolio loss is approximately:

€8.17

Most of the loss comes from German power exposure.

Worst historical portfolio day

The worst observed portfolio outcome occurs on:

19 December 2022

with an estimated portfolio loss of approximately:

€17.76

German power accounts for the majority of this loss.

Main findings

The analysis produces four main conclusions.

First, the energy-market risk factors are highly non-normal and exhibit substantial tail risk.

Second, the portfolio appears diversified in terms of nominal capital allocation, but risk contributions are highly uneven. German electricity dominates the portfolio's extreme losses.

Third, the Gaussian VaR model materially understates extreme tail risk. Historical Simulation performs better at the 99% level because it preserves historically observed extreme movements.

Fourth, even Historical Simulation fails the rolling VaR backtest. Stress losses also substantially exceed conventional VaR estimates.

The results therefore illustrate why energy-market risk management should not rely on a single risk metric or distributional assumption.

Model limitations

The project deliberately uses a simplified portfolio representation.

Important limitations include:

- linear spot-price exposures rather than full derivative revaluation;
- no forward-curve dynamics;
- no options or nonlinear sensitivities;
- no storage valuation;
- fixed physical positions through time;
- fixed USD/EUR conversion;
- no explicit FX risk;
- no liquidity or basis risk;
- no volatility modelling such as GARCH;
- no conditional correlation model.

These simplifications keep the project focused on statistical portfolio-risk modelling rather than full production valuation infrastructure.

Potential extensions

Natural extensions would include:

- conditional volatility models;
- Student-(t) or filtered historical simulation;
- Monte Carlo VaR;
- Expected Shortfall backtesting;
- time-varying correlations;
- forward and futures contracts;
- options and Greeks;
- stochastic forward curves;
- FX risk;
- portfolio risk decomposition;
- marginal and component VaR.

Conclusion

The project demonstrates how real energy-market data can be transformed into a transparent portfolio-risk framework combining:

- exposure modelling;
- P&L construction;
- Value-at-Risk;
- Expected Shortfall;
- rolling model validation;
- formal coverage testing;
- stress testing.

The results show that model assumptions matter materially in energy markets and that empirical validation is essential when risk models are used to support portfolio and trading decisions.